Q11 — Discussion: Why Do Models Hallucinate, and Can They Know When They Are Unsure?

DLS 2026 — Graduate AI-Internals Module

Framing the Question

A student asks why large language models (LLMs) produce confident-sounding falsehoods, and whether a model can represent its own uncertainty. These are two distinct but entangled problems. The first concerns the *generative* tendency of models to emit fluent but ungrounded text. The second concerns *introspective calibration*: whether the model's internal or externalized confidence tracks the probability that its output is correct.

It is useful to separate several mechanisms often lumped under "hallucination." Some errors arise from pretraining objectives that reward plausible continuations rather than truthful ones; Ji et al. [1] provide a taxonomy distinguishing intrinsic from extrinsic hallucination. Others arise from the statistical structure of the training distribution: Kalai and Vempala [2] argue that, under a cross-entropy objective, a nontrivial hallucination rate on rare facts is a mathematically expected consequence of being well-calibrated on the language-modeling task, linking generation errors to the base rate of "singleton" facts seen once in training. Instruction tuning and RLHF can further amplify confident assertion: Ouyang et al. [3] show alignment reshapes output distributions in ways that do not directly optimize factual accuracy, and reward models may inadvertently prefer assertive answers.

On the question of self-knowledge, the evidence is mixed. Kadavath et al. [4] report that models can be trained or prompted to predict whether they "know" an answer, with reasonable calibration on some tasks. Guo et al. [5] established that modern neural networks are frequently *miscalibrated*, typically overconfident, and that post-hoc methods such as temperature scaling help. More recent work on verbalized confidence [6] and on semantic-entropy-based uncertainty estimation [7] and [8] suggests that uncertainty signals exist but must be read out carefully, since token-level probabilities conflate linguistic and epistemic variability. Retrieval augmentation [9] reduces but does not eliminate the problem, since models may still ignore or misweight retrieved evidence.

Five Open Questions for Discussion

Q1. Is hallucination a bug or a predictable equilibrium of the objective? If Kalai and Vempala [2] are right that error on rare facts follows from calibrated density estimation, then eliminating hallucination entirely may require trading away calibration or abandoning pure next-token likelihood. What objective modifications (e.g., abstention-aware losses, selective prediction) could decouple fluency from unwarranted assertion without degrading general capability? Can you design a toy distribution where this trade-off is provably sharp?

Q2. Does a model "know" it is uncertain, or only know how to say so? Kadavath et al. [4] and Lin et al. [6] show models can emit calibrated-looking confidence statements, but

this may reflect learned surface correlations rather than access to an internal epistemic state. What experiment would distinguish genuine internal uncertainty representation (e.g., probes on hidden states, semantic entropy over sampled generations per Kuhn et al. [7]) from mere stylistic hedging learned from training data?

Q3. Which layer of the stack is responsible? Hallucination could originate in pretraining distribution gaps, in the decoding procedure, or in the RLHF reward model's preference for assertiveness [3]. How would you attribute a specific false claim to one of these sources? Can controlled ablations (fixed base model, varying decoding temperature, varying reward model) isolate the dominant contributor for a given task?

Q4. Is uncertainty best measured in probability space or semantic space? Token-level entropy can be high simply because many paraphrases are valid, not because the model is unsure of the fact. Farquhar et al. [8] and Kuhn et al. [7] cluster semantically equivalent generations to recover epistemic uncertainty. When does semantic entropy outperform log-probability-based confidence, and what are its failure modes when the semantic clustering model itself is unreliable?

Q5. Does grounding transfer calibration? Retrieval augmentation [9] supplies external evidence, but a model may still hallucinate when evidence is absent, contradictory, or ignored. Can we build models whose confidence is explicitly conditioned on retrieval support, such that abstention rises when the context is uninformative? How would you evaluate whether post-hoc calibration methods [5] compose with retrieval, or whether grounding and calibration must be co-trained?

Pointers for the Seminar

Students should come prepared to argue a position on at least two of the five questions, grounding claims in the taxonomy of Ji et al. [1] and the empirical self-knowledge results of Kadavath et al. [4]. A productive exercise is to reproduce a small semantic-entropy detector following Farquhar et al. [8] and report where it disagrees with raw log-probability confidence.

References

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